

MARY, Turkmenistan

WMO: 384573

Lat: 37.619N

Lon: 61.897E

Elev: 728

StdP: 14.31

Time Zone: 5.00 (E05)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.7	21.1	5.4	7.5	15.5	12.3	10.6	32.0	20.5	54.5	18.1	48.9	5.0	90	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	107.2	68.4	104.1	67.3	102.1	66.6	71.0	99.1	69.8	97.5	68.7	95.9	10.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	82.9	79.6	59.3	77.5	79.8	57.6	72.7	79.8	35.1	99.2	34.0	96.9	33.1	97.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	16.2	14.4	11.8	110.8	9.0	2.4	5.3	112.5	0.0	113.9	-5.1	115.2	-11.6	116.9		
			10.6	73.9	8.7	1.4	4.3	74.9	-0.8	75.7	-5.6	76.5	-11.9	77.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.4	40.8	43.5	55.5	66.0	78.0	85.4	87.9	84.0	74.7	63.1	50.2	42.5
	DBStd	18.58	10.06	10.39	9.12	7.87	5.42	4.47	4.54	5.06	5.88	8.17	9.55	8.46
	HDD50	955	309	219	49	1	0	0	0	0	0	8	108	261
	HDD65	3037	751	604	316	81	0	0	0	0	4	131	451	699
	CDD50	6214	24	36	219	480	867	1063	1176	1053	740	415	114	27
	CDD65	2821	0	0	21	110	403	613	711	588	294	73	8	0
	CDH74	40086	1	15	297	1325	5025	9227	11146	8459	3600	895	91	5
	CDH80	23984	0	3	89	568	2692	5846	7375	5270	1799	314	28	0
Wind	WSAvg	7.4	7.3	7.9	8.1	8.2	7.9	8.0	8.0	7.4	6.5	6.0	6.0	6.9
Precipitation	PrecAvg	5.9	0.7	0.9	1.1	0.9	0.6	0.2	0.0	0.0	0.0	0.3	0.6	0.6
	PrecMax	9.3	1.4	2.3	1.9	2.4	1.8	1.1	0.3	0.3	0.4	1.1	1.4	1.4
	PrecMin	2.8	0.1	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
	PrecStd	1.8	0.3	0.6	0.4	0.6	0.5	0.2	0.1	0.1	0.1	0.2	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.0	75.4	89.2	97.2	102.5	107.8	111.3	107.9	101.9	94.8	84.2	73.2
		MCWB	51.5	56.5	60.8	64.7	67.0	68.5	69.2	68.6	65.2	63.5	59.9	51.8
	2%	DB	66.0	68.0	81.0	91.4	98.8	104.3	107.9	104.2	97.2	87.6	75.1	66.0
		MCWB	50.0	51.8	58.5	63.6	65.6	67.2	68.8	66.8	63.9	61.0	55.4	51.3
	5%	DB	59.4	62.8	76.7	86.2	96.5	102.3	105.6	102.1	93.4	82.7	69.8	60.5
		MCWB	48.1	51.1	57.2	61.0	64.7	66.4	67.8	66.2	63.3	59.6	54.0	48.7
	10%	DB	54.0	58.7	71.4	82.0	93.1	100.3	102.4	98.9	89.8	78.8	64.7	55.4
		MCWB	45.4	49.0	55.5	60.5	64.1	65.9	67.1	65.4	62.6	58.0	51.5	46.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	57.2	63.5	68.4	70.1	72.0	73.3	71.7	68.4	65.5	61.6	54.3
		MCDB	66.5	72.0	82.1	89.3	93.9	100.0	100.3	99.2	92.8	89.0	79.7	68.0
	2%	WB	51.2	54.2	60.2	65.6	68.3	70.2	71.3	69.7	66.6	62.9	57.8	51.6
		MCDB	63.6	65.6	76.4	85.3	92.0	98.5	100.5	97.0	91.1	83.2	71.1	63.1
	5%	WB	48.7	51.3	58.1	63.3	67.0	68.9	70.1	68.4	65.0	61.0	55.0	49.4
		MCDB	59.4	60.3	73.6	81.0	89.7	96.5	98.9	95.9	89.4	79.3	66.2	59.2
	10%	WB	45.7	49.6	56.3	61.4	65.7	67.6	68.9	67.0	63.5	59.1	52.8	47.0
		MCDB	53.4	57.5	69.9	77.3	86.9	94.6	97.4	94.6	87.0	76.9	63.5	54.3
Mean Daily Temperature Range	5% DB	MDBR	17.8	17.9	20.7	22.5	24.6	26.3	27.5	27.7	27.4	25.1	20.8	17.3
		MCDBR	26.5	26.0	29.7	31.9	29.5	29.6	30.5	30.2	32.0	30.9	28.1	26.1
		MCWBR	14.3	14.2	12.3	12.5	9.9	9.3	9.9	10.4	12.1	13.8	14.2	14.3
	5% WB	MCDBR	25.0	22.7	27.2	26.8	25.8	26.7	27.8	27.6	28.4	27.8	25.4	23.9
		MCWBR	14.2	13.1	12.3	12.1	9.4	9.2	9.8	10.0	11.7	12.8	13.4	14.0
Clear-Sky Solar Irradiance	taub	0.391	0.437	0.508	0.540	0.542	0.548	0.532	0.487	0.459	0.488	0.448	0.398	
	taud	2.086	1.971	1.827	1.799	1.817	1.796	1.837	1.925	1.963	1.874	1.953	2.061	
	Ebn at Noon	239	243	237	239	241	239	242	250	249	222	214	225	
	Edh at Noon	38	49	62	67	67	69	65	58	53	53	43	37	
All-Sky Solar Radiation	RadAvg	712	1018	1379	1820	2249	2544	2506	2317	1912	1343	873	658	
	RadStd	60	104	107	154	82	77	55	57	47	77	77	75	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	-1.04	-0.83	N/A	N/A	N/A	N/A

Nomenclature: See separate page